Evidence-Adaptive Decoding for Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation (RAG) usually concatenates several retrieved passages into one flat context before decoding, which is fragile when the passages disagree, differ in relevance, or only partially answer the query. We study Evidence-Adaptive Decoding for RAG (EAD-RAG), a training-free decoding-time policy that treats the retrieved set as an ensemble and arbitrates among document-conditioned candidates using five non-oracle signals: cross-document agreement, retrieval-confidence calibration, token-level entropy and agreement gating, reference-trustable logit shaping, and source-aware verifier reranking. The policy never consults gold labels. We pair EAD-RAG with a same-slice protocol over three public benchmark families—ALCE, RAGTruth, and FActScore—with coverage controls (answer length, response ratio, citation recall, atomic-fact count) that reject factuality gains from deletion or abstention. We report a preliminary, underpowered pilot (at most two paired trials per cell; hosted proxy scorers): in it EAD-RAG does not outperform standard RAG, scoring 0.60 versus 0.65 on ALCE, 0.35 versus 0.72 on FActScore, and 0.82 versus 0.92 on RAGTruth; every paired comparison is underpowered, so we make no significance or method claim. Our two reportable contributions are the method design and an evaluation-integrity analysis: the full matrix repeatedly failed closed on candidate-quality and comparability gates—output-token truncation, citation-dominated and repeated-citation candidates, and hosted-scorer non-comparability on ALCE—which is itself a finding about how hard fair, same-slice RAG-decoding evaluation is.

1 Introduction

Retrieval-augmented generation grounds a language model on retrieved passages so that generation can rely on external evidence rather than parametric memory alone (Lewis et al., 2020). The dominant recipe is deceptively simple: retrieve the top-k passages, concatenate them into a single prompt, and decode as if they formed one coherent source. Two lines of evidence show that this recipe does not, by itself, make generation faithful. Citation-grounded evaluation of long-form answers reveals that many statements a model emits are not actually supported by the passages it cites (Gao et al., 2023). Span-level annotation of RAG outputs shows that unsupported and contradictory spans persist even when the correct source content is present in the context (Niu et al., 2024), and atomic-fact scoring exposes the same problem in long-form factual writing (Min et al., 2023). Retrieval delivers the evidence; the decoder still has to use it.

The flat-context recipe is brittle precisely because it discards structure that was available at retrieval time. Retrieved passages routinely disagree with one another, vary widely in relevance, and interleave supporting content with distractors. Once they are concatenated, a decoder has no explicit notion of *which* document a token is grounded in, so a claim that one passage supports but others contradict can be emitted as if it were consensus. The result is fluent text whose support is uneven, and it is exactly this failure that citation-grounded and span-level benchmarks penalize.

A natural response is to intervene at decoding time without retraining the model. Training-free decoding methods show that logits can be reshaped using external references (Shi et al., 2024), retrieval signals (Nguyen et al., 2025), or model-confidence contrasts (Li et al., 2026), and mechanistic studies describe how retrieved evidence and citations are actually used inside the network during generation (Sun et al., 2025; van Dort and Heuss, 2026). These methods, however, mostly optimize a generic notion of truthfulness or a single confidence contrast; they do not directly address the RAG-specific problem of *conflicting documents with uneven trust*. In parallel, post-hoc detectors flag unsupported spans after the fact (Kovacs and Recski, 2025; Manakul et al., 2023), but a detector does not define how token-level generation should adapt when the evidence disagrees. The gap this paper targets is a decoding policy that arbitrates among document-conditioned candidates using only signals that are available without gold labels.

Method insight. We propose Evidence-Adaptive Decoding for RAG (EAD-RAG): rather than collapsing retrieved passages into one flat context, the decoder treats them as an ensemble and composes five non-oracle signals—cross-document agreement, retrieval-confidence calibration, entropy and agreement gating, reference-trustable shaping, and source-aware verifier reranking—into a single policy whose inputs are strictly separated from evaluation labels. The design question is whether these decoding-time signals can reduce unsupported generation *while preserving* answer coverage and usefulness, since a decoder can trivially inflate a factuality score by deleting content or abstaining.

Honest preview. We are explicit that this paper reports a *preliminary* study. Only underpowered diagnostic pilots have produced paper-facing rows; the full public-benchmark matrix has not. In a paired pilot with a single trial per cell and hosted proxy scorers, EAD-RAG scores below standard RAG on all three families—0.60 versus 0.65 on ALCE, 0.35 versus 0.72 on FActScore, and 0.82 versus 0.92 on RAGTruth. Because each cell has at most two paired trials, every comparison is underpowered; we report no p-values and make *no* method or significance claim. What we can report honestly is (i) the method design and (ii) an evaluation-integrity finding: repeated attempts to run the full matrix failed closed on candidate-quality and comparability gates rather than producing valid rows.

Contributions.

1. A training-free, non-oracle evidence-adaptive decoding policy for RAG that arbitrates among document-conditioned candidates instead of trusting one flat context, with an explicit label-hidden policy boundary (Section 3).
2. A same-slice evaluation protocol over three independent public benchmark families with coverage and usefulness controls that are designed to reject deletion-only and abstention-only "gains" (Section 4).
3. A preliminary, honestly-labeled pilot showing that EAD-RAG does not currently outperform standard RAG, together with a paired-significance statement that declines to claim any effect from underpowered data (Section 5).
4. A candidate-quality and comparability failure analysis—output-token truncation, citation-dominated candidates, repeated-citation sequences, and hosted-scorer non-comparability—that documents concretely why fair same-slice RAG-decoding evaluation is hard (Section 6).

Scope. We do not claim that EAD-RAG improves faithfulness. The evidence here is pilot-scale, and the method claim is deferred to a fully powered matrix. The paper's positive content is the policy design and the evaluation-integrity analysis.

2 Related Work

Citation-grounded RAG evaluation and hallucination benchmarks. ALCE frames citation-grounded answering as a benchmark, scoring whether statements are supported by cited passages across ASQA, QAMPARI, and ELI5-style tasks (Gao et al., 2023). RAGTruth contributes word-level hallucination labels across question answering, data-to-text, and summarization, and demonstrates that unsupported and contradictory spans survive even when the source is present in context (Niu et al., 2024). FActScore decomposes long-form generations into atomic facts and scores each for support, giving a factuality lens that is independent of citation formatting (Min et al., 2023). These families are complementary in what they penalize, which is why we evaluate on all three rather than a single metric. Their shared limitation for our purposes is that they measure final outputs and do not prescribe how decoding should change; moreover, running several methods over the *same* slice under a *single* comparable scorer is itself non-trivial, a difficulty that becomes the empirical centerpiece of this paper.

Training-free and confidence-aware decoding. Reference Trustable Decoding (RTD) reshapes generation toward trusted references using a non-parametric datastore, without fine-tuning (Shi et al., 2024). Retrieval-Augmented Decoding (RAD) injects retrieval signals at decoding time to improve truthfulness in open-ended generation (Nguyen et al., 2025), and Dual-Confidence Contrastive Decoding (DCCD) contrasts confidence estimates to steer RAG generation (Li et al., 2026). Token-level analyses such as CORTEX compare internal representations to localize where a RAG generation goes unsupported (Furumai et al., 2026). EAD-RAG borrows the training-free, logit-shaping spirit of RTD but differs in target: it treats retrieved documents as an ensemble and arbitrates among document-conditioned candidates with retrieval-calibrated trust and entropy gating, rather than optimizing a generic truthfulness objective or a single confidence contrast. Whether this document-ensemble arbitration is separable from RTD and the nearest confidence-decoding baselines under matched same-slice evaluation is a question we pose but do not resolve here; establishing that separation is a required close-baseline comparison that the full matrix has not yet delivered.

Grounding mechanisms and attribution. A complementary line studies *how* models ground generations. ReDeEP decouples a model's use of external context from its parametric prior and uses the imbalance to detect hallucination in RAG (Sun et al., 2025). Span-level grounding extends detection beyond documents to code and tool output (Kovacs et al., 2026), and mechanistic work traces how attribution and citation behavior arise during generation (van Dort and Heuss, 2026). These studies motivate the non-oracle signals EAD-RAG uses—agreement, source support, and span-level grounding—but they are analytic or detection-oriented: they explain or flag grounding failures without specifying a decoding policy that acts on them token by token.

Detectors, sampling, and classic baselines. Practical RAG hallucination detectors such as LettuceDetect show sustained demand for post-generation checking (Kovacs and Recski, 2025), and SelfCheckGPT uses sampled-output consistency as a zero-resource hallucination signal (Manakul et al., 2023). Self-consistency aggregates multiple sampled generations and marginalizes over them (Wang et al., 2022), and the original RAG architecture couples parametric generation with non-parametric retrieval (Lewis et al., 2020). These methods anchor our baselines: standard RAG and self-consistency are the status-quo comparisons, and a verifier-reranking baseline captures what a post-hoc detector alone can achieve. The gap they leave is prevention at decode time: detection and sampling do not by themselves keep unsupported tokens from being generated when the retrieved evidence is contested.

3 Method: EAD-RAG

EAD-RAG is a training-free decoding-time policy. It performs no gradient updates, supervised fine-tuning, or preference optimization; it wraps a fixed base model and a fixed retriever. The design principle is that the retrieved set carries structure—provenance, retrieval rank, and cross-document (dis)agreement—that the flat-context recipe throws away, and that a decoder can recover and act on this structure without ever seeing gold labels.

3.1 Setup and notation

For a query q, the retriever returns passages $D = \{d_1, \dots, d_k\}$ with retrieval scores $s_1, \dots, s_k$. Let p_θ denote the fixed base model. The flat-context baseline decodes from $p_\theta(\cdot \mid q, d_1 \oplus \cdot \oplus d_k)$, where $\oplus$ is concatenation. EAD-RAG instead maintains *document-conditioned* next-token distributions

$$p_i(\cdot) = p_\theta(\cdot \mid q, d_i, y_{<t}), \qquad i = 1, \ldots, k, \quad (1)$$

where $y_{<t}$ is the partially generated answer. The policy operates on the tuple of permitted inputs

$$\Phi = \big(q,\ D,\ \{s_i\},\ \{p_i\},\ \text{candidate text},\ v(\cdot)\big), \quad (2)$$

where $v(\cdot)$ is a source-aware verifier support score. Crucially, Φ excludes every evaluation field (Section 3.4).

3.2 The five components

(1) Document-ensemble decoding. Rather than decoding once over the concatenation, EAD-RAG forms a trust-weighted mixture over document-conditioned distributions,

$$m_t(\cdot) = \sum_{i=1}^{k} w_i\, p_i(\cdot), \quad (3)$$

so that a token supported by one passage but contradicted by others is not silently emitted as consensus. The mixture makes cross-document disagreement an explicit, actionable quantity rather than an artifact hidden inside a single prompt.

(2) Retrieval-confidence calibration. The mixture weights are a calibrated function of retrieval score,

$$w_i = \frac{\exp(s_i/\tau)}{\sum_{j=1}^{k} \exp(s_j/\tau)}, \quad (4)$$

with temperature τ controlling how sharply high-ranked passages dominate. Calibration prevents a low-relevance passage from contributing as much as a strongly retrieved one, limiting the influence of distractors.

(3) Entropy and agreement gating. EAD-RAG adapts only where the evidence is contested. Let $H_t = H(m_t)$ be the entropy of the mixture and let $A_t \in [0,1]$ measure cross-document agreement (the normalized concentration of the top-token vote across $\{p_i\}$). The gate

$$g_t = \mathbb{1}[\, H_t > \theta_H \ \text{ or } \ A_t < \theta_A \,] \quad (5)$$

fires when the model is uncertain or the documents disagree, concentrating adaptation on exactly those tokens and otherwise deferring to ordinary decoding.

(4) Reference-trustable shaping. Following the training-free spirit of RTD (Shi et al., 2024), at gated tokens the policy adds a reference-support term $r_t(\cdot)$ derived from spans that the retrieved passages support, producing shaped scores

$$\ell_t(\cdot) = \log m_t(\cdot) + \lambda\, g_t\, r_t(\cdot), \quad (6)$$

with strength λ. Shaping is applied only where the gate fires, so it biases generation toward supported continuations without overriding confident, uncontested tokens.

(5) Source-aware verifier reranking. After candidate answers are produced, a verifier that is aware of which passage a span is grounded in scores each candidate for source support, and the final answer is

$$y^\star = \arg\max_{y \in \mathcal{Y}}\ v(y; D), \quad (7)$$

so that support, not surface fluency, drives selection. If the best verifier score falls below a confidence floor, $\max_y v(y; D) < \theta_V$, the policy falls back to standard RAG or to a conservative shortened answer rather than committing to a weakly supported candidate.

3.3 Decision rule

One generation episode proceeds as follows.

1. Retrieve or load released evidence D and compute calibrated weights $\{w_i\}$.
2. Decode document-conditioned candidates, forming the mixture m_t and the agreement statistic A_t at each step.
3. At each token, evaluate the gate g_t; if it fires, apply reference-trustable shaping, otherwise decode from m_t unchanged.
4. Collect the candidate set $\mathcal{Y}$ and rerank with the source-aware verifier.
5. Emit $y^\star$, or fall back / abstain when the verifier floor θ_V is not met.
6. Log all policy inputs and outputs for audit.

The controllers $\tau, \theta_H, \theta_A, \lambda, \theta_V$ are policy hyperparameters and are never tuned on visible evaluation labels; they are set from the permitted inputs and a held-out development split only.

3.4 Non-oracle policy boundary

The policy may consult retrieved passages and their retrieval scores or ranks, per-document logprobs and entropy, cross-document agreement, verifier support scores computed from sources and candidates, and candidate-level metadata such as answer length and citation coverage. It may *not* consult RAGTruth hallucination labels, ALCE gold answers, FActScore labels, oracle-reranked retrieval as a primary condition, or any target field. This boundary is what makes the setting realistic: at deployment there are no gold labels, so a policy that quietly depends on them measures nothing useful. We enforce the boundary with a benchmark-side oracle-leakage check that materializes the exact policy inputs and asserts that no evaluation field is present before any scored run proceeds (Section 4).

3.5 Cost and practicality

The document-ensemble step evaluates up to k document-conditioned distributions per token, which is the main overhead relative to a single flat-context pass. Two design choices keep this tractable. First, the entropy and agreement gate restricts the more expensive shaping computation to tokens where the evidence is actually contested, so the per-token cost approaches the flat-context baseline on uncontested spans. Second, the candidate pool that the verifier reranks is small and bounded, so reranking is a fixed additive cost per query rather than a search over an exponential space. Because the policy is training-free, it introduces no offline training cost and can wrap an existing deployed model and retriever; its budget is entirely at inference time, which is what the run-stage token and cost counters record. These properties matter for the honest comparison we set up next: any support gain the method might show has to be weighed against this inference overhead, and a gain that appears only under much longer generation budgets is not a fair improvement.

4 Experimental Setup

Benchmark provenance. We evaluate on three independent public benchmark families, summarized in Table 1. They are chosen because they penalize different failure modes—

Family	Planned tasks	Primary metric	Access
ALCE	180	Citation support	Public repo
RAGTruth	240	Source faithfulness	Public repo
FActScore	183	Atomic factual precision	Public repo
Total	**603**	Three families	Research use

Table 1: Benchmark provenance. Three independent public families with label-hidden policy inputs; "Planned tasks" is the intended same-slice matrix size per method condition. The full matrix has not produced valid paper-facing rows, so the reported pilot uses at most two paired trials per cell.

citation support, span-level faithfulness, and atomic factual precision—and because their labels can be held strictly on the scoring side of the non-oracle boundary. Planned same-slice sizes are 180 ALCE tasks, 240 RAGTruth examples, and 183 FActScore entities, for a planned matrix of 603 unique instances per method condition. The pilots reported below use a much smaller slice (at most two paired trials per cell) because the full matrix has not yet produced valid paper-facing rows.

Backend. Candidate generation uses a hosted instruction-tuned language model accessed through a chat/responses API, held fixed across every method and baseline so that differences are attributable to the decoding policy rather than the backbone. A separate small local open-weights inference smoke was also run through a batched inference server; it is likewise underpowered and mixed and is not used to support any claim. Scoring uses hosted proxy scorers: a citation-support proxy for ALCE, a factual-precision proxy for FActScore, and a source-faithfulness scorer for RAGTruth drawn from the released-label-calibrated family. We name no vendor model identifiers.

Baselines. The intended comparison set is standard (flat-context) RAG, self-consistency (Wang et al., 2022), verifier-reranking-only, Reference Trustable Decoding (Shi et al., 2024), and the nearest executable confidence-decoding baseline (Nguyen et al., 2025; Li et al., 2026). The verifier-reranking-only baseline is required *before* any token-level claim, because if reranking alone matches the full policy under coverage controls, the decoding components add nothing. The pilot reported here exercises standard RAG, verifier-reranking-only, and EAD-RAG; RTD and the confidence-decoding line are identified close baselines whose faithful same-slice reproduction is outstanding.

Metrics and coverage controls. Each family contributes a primary support or factuality score in $[0, 1]$. Every such score is paired with coverage and usefulness controls—answer length, response ratio, citation recall, and number of atomic facts per response—so that a gain produced by deleting content or abstaining is detected rather than credited. A method that improves support only by shortening or refusing answers is not counted as an improvement.

Budget, seeds, and stopping rules. Runs record a manifest, status, progress log, and results file, together with token and cost counters and started/ended timestamps, and check a STOP file before each expensive call. The pilots use one to two paired trials per cell; with a single pair per cell, paired tests are underpowered by construction and we report deltas without p-values. Automated early stops fire if benchmark labels appear in policy inputs, if task slices differ across conditions, if output rows lack a method, task id, prediction, or score field, if the cost or token budget is exceeded, or if repeated backend failures persist after retry. These stops are what caused the full matrix to fail closed rather than emit questionable rows (Section 6).

5 Results

No method claim is supported. We report the paired pilot in Table 2 exactly as it stands: underpowered, hosted/proxy-scored, and not eligible to support any positive claim. In this pilot EAD-RAG does not outperform standard RAG. On ALCE it scores 0.60 against standard RAG's 0.65; on FActScore 0.35 against 0.72; and on RAGTruth 0.82 against 0.92. The verifier-reranking-only baseline is competitive with or above standard RAG on RAGTruth

Method	ALCE	FActScore	RAGTruth
Standard RAG	0.65	0.72	0.92
Verifier-rerank only	0.40	0.75	0.98
EAD-RAG (ours)	0.60	0.35	0.82
Δ (EAD−Std)	−0.05	−0.37	−0.10

Table 2: **Preliminary, underpowered pilot—not for claims.** One paired trial per cell (n=1); hosted proxy scorers; scores in $[0, 1]$. The ALCE column is a hosted citation-support proxy explicitly marked non-comparable to the released ALCE citation metric, and the FActScore column is an uncalibrated hosted factual-precision proxy; only RAGTruth is scored by a released-label-calibrated scorer. In this pilot EAD-RAG does not beat standard RAG on any family; all paired comparisons are underpowered, so no significance or method claim is made.

(0.98 versus 0.92), the only pilot column scored by a released-label-calibrated scorer, and on the uncalibrated hosted FActScore proxy (0.75), while being lower on the hosted ALCE proxy (0.40); this is consistent with the coverage-artifact reading in Section 6 and does not by itself establish that reranking helps.

The single most important thing to read from Table 2 is what it is *not*: it is not a benchmark result. Each cell has one paired trial; the ALCE column is scored under a proxy that the project marks non-comparable to the released ALCE citation metric; and a separate local inference smoke with two trials per cell was likewise mixed, with EAD-RAG not reliably beating standard RAG. We therefore treat the numbers as diagnostic only and move to the analysis that we regard as the paper's real empirical contribution.

6 Analysis: Why the Matrix Failed Closed

The central empirical finding of this study is not a score but a process observation: repeated attempts to run a fair, same-slice RAG-decoding matrix did not produce valid paper-facing rows because the generated candidates failed candidate-quality and comparability gates. We view this as substantive. Fair RAG-decoding evaluation requires that every method produce well-formed, comparable answers on the same inputs under one comparable scorer, and each of the following failure modes breaks that requirement in a concrete, reproducible way.

Paired-significance statement. We first make the statistical scope explicit. In the paired pilot with a single pair per cell, EAD-RAG's per-family deltas versus standard RAG are -0.05 (ALCE), -0.37 (FActScore), and -0.10 (RAGTruth). With one paired observation per cell the paired test is underpowered by construction; we report no p-values and make no significance claim, in either direction. The two-trial local smoke is equally underpowered. No comparison in this paper meets the sample-size bar for a significance statement.

Failure mode 1: output-token truncation. Several candidate builds hit the maximum-output-token cap and were cut off mid-answer. Truncated candidates corrupt both the primary score and the coverage controls at once: a half-finished answer can look "more precise" simply because it stopped before making a claim. The iterative repair history is visible in the run bundle names, which progress through stop-repair and answer-first variants, and some standard-RAG ALCE cells errored on every trial rather than completing. Truncation implies that a decoding-time comparison must fix and verify the generation length budget before scoring, or length becomes an uncontrolled confound.

Failure mode 2: citation-dominated candidates. On ALCE, some candidates degenerated into strings that were mostly bracketed citation markers with little answer text. Such candidates can score unpredictably under a citation-support proxy while carrying almost no information, and they are not comparable to a full-sentence answer from another method. This is why several ALCE cells register very low proxy scores despite completing. The implication is that citation coverage must be reported jointly with answer content; a citation-support number in isolation is not a faithfulness measurement.

Failure mode 3: repeated-citation sequences. A related decoding degeneracy is repetition of the same citation token in sequence. Repeated-citation loops are a generation pathology, not evidence use, and they break comparability because the scorer is fed a degenerate string rather than an answer. Detecting and rejecting such sequences is a prerequisite for a clean same-slice comparison, and doing so is one reason the guarded matrix runs terminated without paper-facing rows.

Failure mode 4: hosted-scorer non-comparability on ALCE. The most consequential gate is comparability of the scorer itself. The ALCE citation support used in the pilot is a hosted proxy that the project could not calibrate to the released ALCE citation metric; every ALCE row is therefore marked non-comparable and fails closed on calibration. A non-comparable scorer means cross-method ALCE numbers cannot be placed on one axis, so the ALCE column of Table 2 is presented only as a proxy and never as a claim. Of the three pilot columns, only the RAGTruth scorer is drawn from a released-label-calibrated family; the hosted FActScore column is an uncalibrated factual-precision proxy and is therefore diagnostic only as well. The released-label FActScore atomic-support scorer—the one FActScore metric we regard as fully calibrated—does not appear in this hosted pilot at all, but in the separate local smoke discussed next, where it exposes a different problem.

The local smoke reinforces the candidate-quality reading. A separate small local open-weights inference smoke (two trials per cell) provides a sharper version of the same lesson. Under the released-label FActScore atomic support scorer—the scorer we consider most trustworthy because it is calibrated to released labels—every FActScore candidate in that smoke scored 0.00 across standard RAG, verifier-reranking, and EAD-RAG alike. A uniform zero under a trustworthy scorer is not a method comparison; it is a signal that the local candidates were not well formed enough to contain scorable, supported atomic facts. That the failure is shared across all three conditions is exactly the point: it is a property of the generation and scoring apparatus, not of any one decoder, and it would be indefensible to read a method ranking off cells where every method scores zero. The RAGTruth ordering in the same smoke flipped relative to the hosted pilot, which is expected of two-trial

Family	Method	Chars	Cites	Facts	Uns.
ALCE	Standard	305	2	–	1
ALCE	Verifier	200	1	–	1
ALCE	EAD-RAG	303	2	–	1
RAGTruth	Standard	686	–	–	1
RAGTruth	Verifier	482	–	–	0
RAGTruth	EAD-RAG	339	–	–	1
FActScore	Standard	732	–	6	4
FActScore	Verifier	219	–	2	1
FActScore	EAD-RAG	216	–	2	3

Table 3: Coverage and candidate-quality diagnostics from the paired pilot. On RAGTruth and FActScore the pilot EAD-RAG configuration kept substantially fewer characters than the baselines, so its scores reflect a deletion artifact rather than an evidence-aware decision; response ratio was 1.0 for every row.

cells and is why we say the pilots are mixed rather than convergent.

Coverage confounds the pilot ordering. Even where scores exist, the coverage controls show why the pilot ordering must not be read as a method effect. Table 3 reports the per-condition answer length, citations, atomic facts, and unsupported spans from the pilot diagnosis. On ALCE the three methods are nearly indistinguishable in length and span count, so the small score differences carry little signal. On RAGTruth and FActScore the picture is a coverage *artifact*: the pilot EAD-RAG branch kept only 339 characters on RAGTruth against the verifier's 482, and only the first two biography sentences (216 characters) on FActScore, where that family has no source document for an evidence-aware decision. A method that scores by keeping fewer characters is exactly what the coverage controls exist to reject, and here they flag the pilot EAD-RAG configuration rather than vindicate it.

A pre-registration-style de-risk signal. Before the pilot, a high-precision evidence gate was screened on 30 labeled RAGTruth rows as a zero-cost sanity check. It moved the unsupported-character rate from 0.122246 to 0.087310 (a change of -0.034936, lower being better) at zero API cost. This is a research-stage screen, not a benchmark result, and we report it only to show that the high-precision idea produced a directional signal worth a fully powered test—not as evidence that EAD-RAG works.

What the failure analysis implies. Taken together, the four gates and the coverage confound argue that the hard part of RAG-decoding evaluation is not inventing a decoder but building an apparatus in which several decoders produce well-formed, length-controlled, comparably-scored answers on identical inputs. Every gate that fired is a place where a less careful pipeline would have emitted a number that looked like a result. We prefer to report the closed gates.

7 Limitations

This is a preliminary study and its limitations are foundational rather than peripheral. First, the evidence is pilot-scale: at most two paired trials per cell, far below the planned 603-instance matrix, so no effect is estimable and no method claim is made. Second, the ALCE column is scored by a hosted proxy that is non-comparable to the released ALCE citation metric, so ALCE numbers are diagnostic only. Third, the method is training-free and therefore bounded by the fixed base model and retriever, and its verifier component can propagate its own errors. Fourth, the pilot EAD-RAG configuration exhibited a deletion artifact on RAGTruth and FActScore, which the coverage controls correctly flagged; a valid method run must first remove that artifact. Finally, required close baselines (Reference Trustable Decoding and the nearest confidence-decoding line) have not been reproduced on the same slice, so the novelty separation the method assumes remains untested. We state these plainly rather than softening them.

8 Conclusion

We presented EAD-RAG, a training-free evidence-adaptive decoding policy that treats retrieved passages as an ensemble and arbitrates among document-conditioned candidates using cross-document agreement, retrieval-confidence calibration, entropy and agreement gating, reference-trustable shaping, and source-aware verifier reranking, all behind a strict non-oracle boundary. We are candid about the evidence: in an underpowered pilot EAD-RAG does not outperform standard RAG, and we make no method or significance claim. The contribution we do stand behind is twofold—the policy design, and an evaluation-integrity analysis showing that a fair, same-slice RAG-decoding matrix repeatedly failed closed on candidate-quality and comparability gates. Building the apparatus that survives those gates is, in our reading, the prerequisite for any credible RAG-decoding result, and it is the next step this work sets up.

Reproducibility and Ethics

All benchmark labels are held on the scoring side of the non-oracle boundary and are never exposed to the decoding policy; a benchmark-side leakage check enforces this before any scored run. Every run records a manifest, status, progress log, results file, and token/cost counters, and honors a stop-file cancellation contract. We report closed gates and underpowered pilots as such and decline to present proxy or non-comparable scores as findings, consistent with an anti-fabrication discipline that treats a fail-closed blocker as the correct output when evidence is insufficient. The three benchmark families are public research releases used under their respective repository terms.

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